SECTION 1

Clustering Foundations and K-Means

- **Unsupervised grouping:** Clustering assigns instances to groups without labels, useful for exploration, segmentation, and preprocessing.
- **Applications:** Customer segments, image color quantization, search by similarity, semi-supervised label propagation, and outlier screening.
- **`KMeans`:** Iterates assign to nearest centroid and recompute means until convergence; `n_clusters=k` must be supplied.
- **Voronoi regions:** Decision boundaries are perpendicular bisectors between centroids; `transform` yields distances to all centers.
- **Initialization:** `n_init` runs multiple seeds; K-Means++ spreads initial centroids; inertia scores the best run.
- **Variants:** Elkan accelerates standard K-Means; `MiniBatchKMeans` updates with subsets for large-scale data.
- **Choosing k:** Elbow on inertia or silhouette score balances cohesion versus separation; diagrams inspect cluster quality.

SECTION 2

K-Means Limits and Clustering Applications

- **Assumptions:** K-Means favors roughly spherical, similar-sized blobs; elongated or uneven clusters break the method.
- **Scaling:** Features should be standardized so one axis does not dominate distance.
- **Image segmentation:** Pixels clustered in RGB space and replaced by centroid colors simplify images.
- **Preprocessing pipeline:** K-Means distances as features before logistic regression on digits can cut error versus raw pixels.
- **Semi-supervised learning:** Label centroid-near exemplars, propagate to clusters, optionally limit to closest fraction per cluster for cleaner pseudo-labels.

SECTION 3

DBSCAN and Other Clustering Algorithms

- **Density-based clusters:** DBSCAN marks core points with enough neighbors within ε, expands clusters, and labels sparse points as noise (-1).
- **Arbitrary shapes:** Works when clusters are dense islands separated by low density; `eps` and `min_samples` are critical.
- **Prediction workaround:** No `predict`; train `KNeighborsClassifier` on core samples to assign new points and optional distance-based anomalies.
- **Alternatives surveyed:** Agglomerative, BIRCH, mean-shift, affinity propagation, and spectral clustering differ in scalability, shape assumptions, and need for connectivity graphs.

SECTION 4

Gaussian Mixture Models

- **Generative mixtures:** Data assumed drawn from k Gaussians with weights ϕ, means μ, and covariances Σ; cluster identity is latent.
- **`GaussianMixture`:** EM alternates responsibility (soft assignments) and parameter updates; supports `predict` and `predict_proba`.
- **`covariance_type`:** `full`, `tied`, `diag`, `spherical` trade flexibility against parameters and stability in high dimensions.
- **Sampling and density:** `sample` draws new points; `score_samples` gives log probability density for each observation.
- **Model selection:** BIC and AIC penalize complexity when choosing k; `BayesianGaussianMixture` can shrink unused components toward zero weight.

SECTION 5

Anomaly and Novelty Detection

- **Low-density outliers:** Points with small mixture density or poor fit are flagged as anomalies; thresholds set contamination rates (e.g., 4% in manufacturing).
- **Novelty vs anomaly:** Novelty detection trains on clean data; anomaly detection tolerates mixed training sets.
- **Robust fitting:** Refit after removing extreme points or use EllipticEnvelope for Gaussian inliers amid contamination.
- **Other detectors:** Isolation Forest isolates points quickly in random trees; LOF compares local density to neighbors; one-class SVM separates training data from the origin in kernel space; PCA reconstruction error flags hard-to-compress instances.